

MODELING INTERFACE EVOLUTION THROUGH TOPOLOGY CHANGES



DIFFUSE-INTERFACE REGULARIZATION

Filament collapse is regularized
by a diffuse layer of width ϵ



Quasi-incompressible Cahn–Hilliard fluids and topological transitions

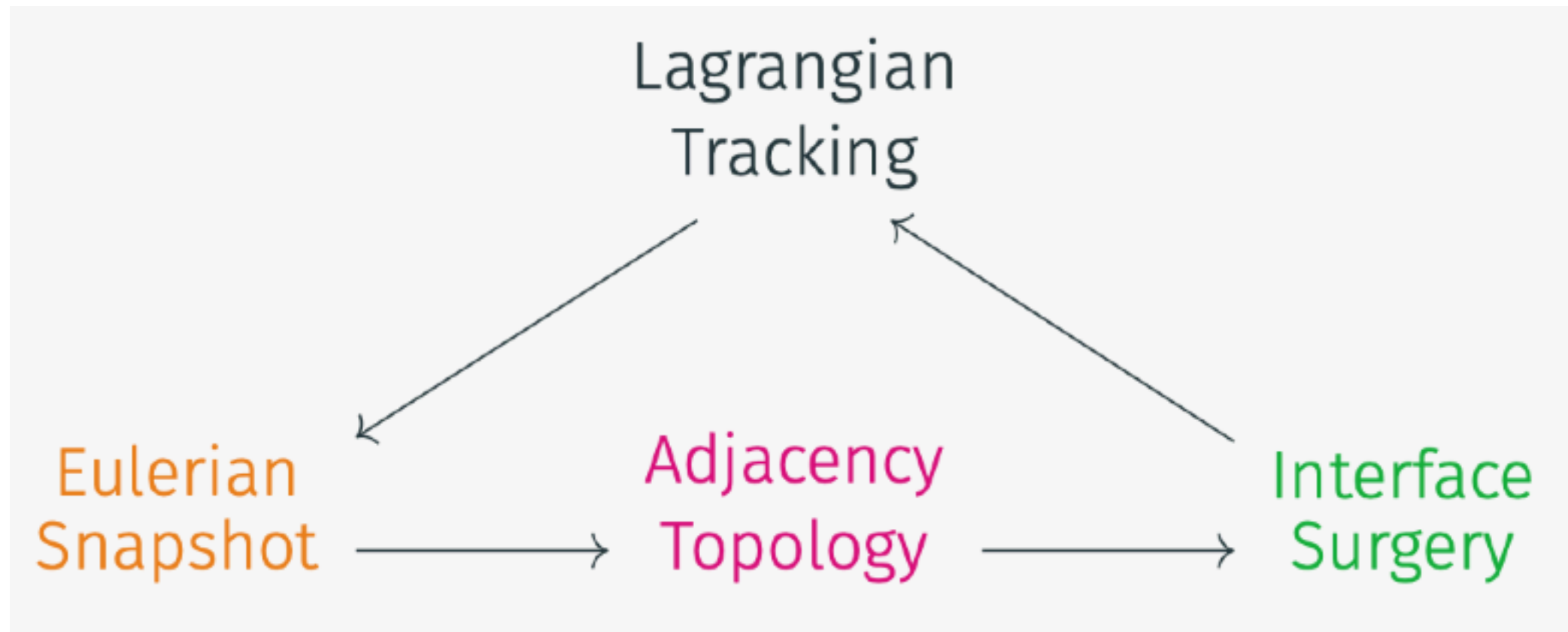
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"...it is natural to conclude that the ideal numerical approach should be based on the sharp interface model and then use 'jump conditions' to transit the sharp interface through a topology change."





Sharp Interface $\longrightarrow$ Diffuse Layer

Topological
singularity

Regularized
transition



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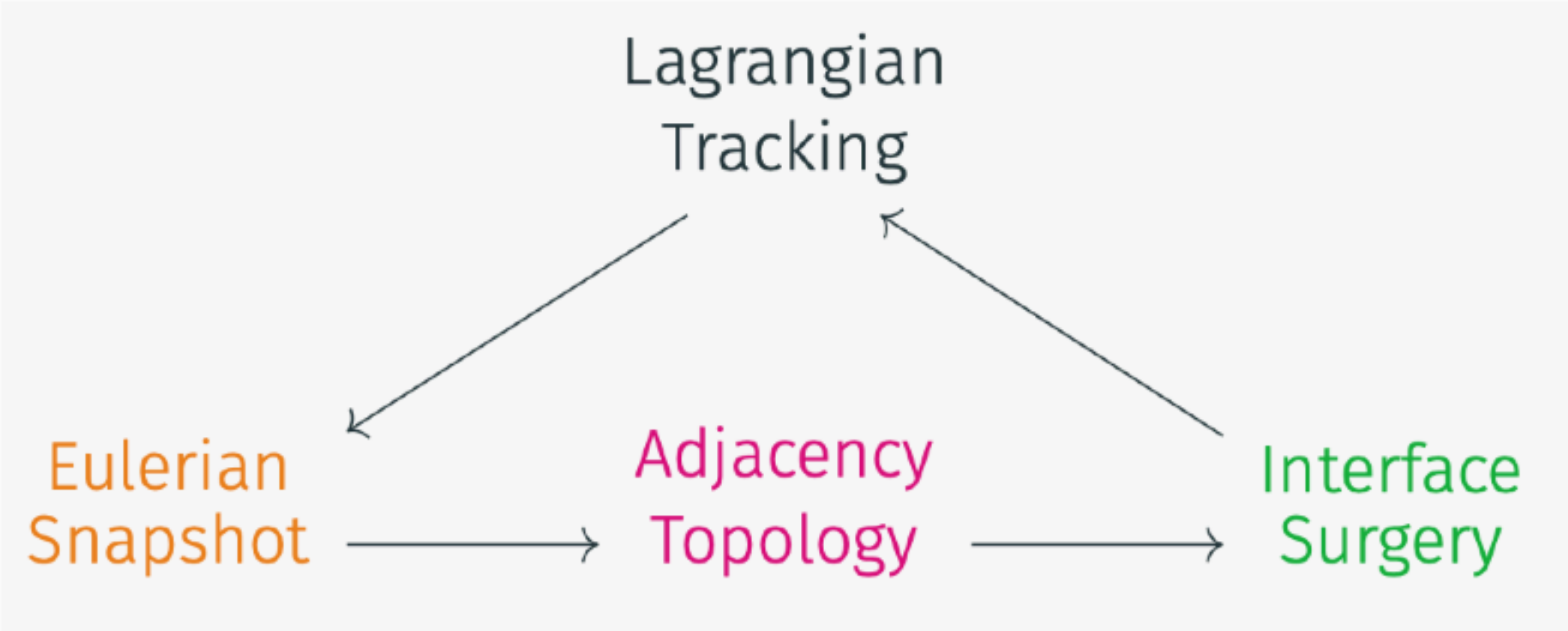
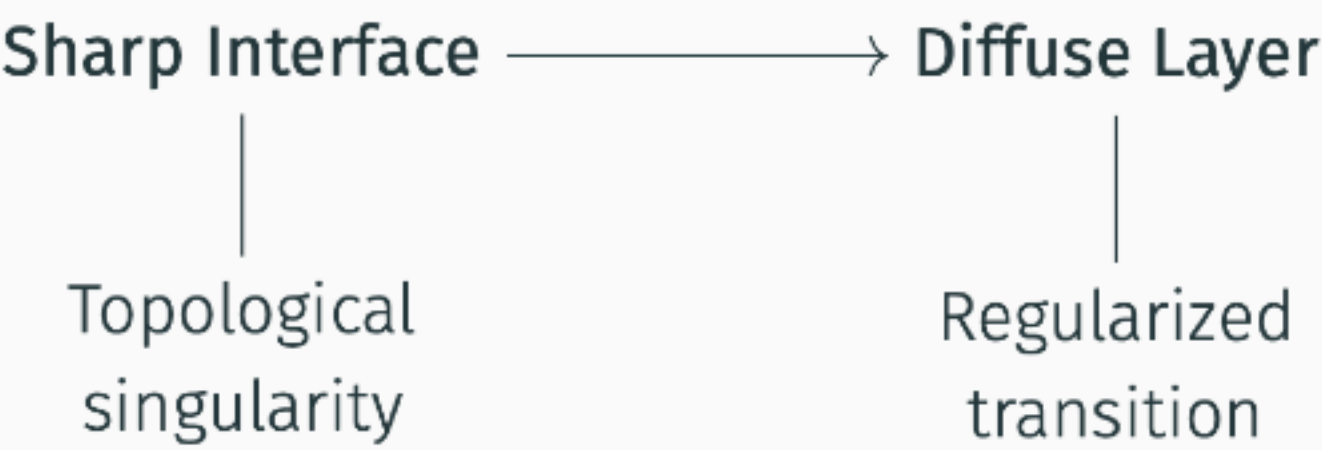
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MICROSCALE TOPOLOGICAL SURGERY (MTS)

INTERMITTENT TOPOLOGICAL CHANGES VIA GEOMETRIC REGULARIZATION AT MICROSCALE

